Lecture-18 - Continous State MDP & Model Simulation

- Discretization
- Models/ Simulation
- Fitted value iteration

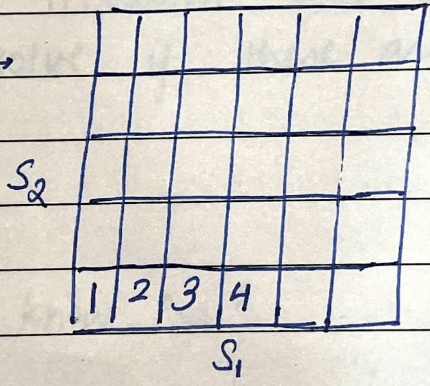
$$\pi^*(s) = \arg\max_a \underbrace{\sum_{s'} P_{sa}(s')\, V^*(s')}_{E_{s' \sim P_{sa}}[V^*(s')]}$$

Value iteration:

$$V(s) = R(s) + \gamma \max\left(\sum_{s'} P_{sa}(s')\, V(s')\right)$$

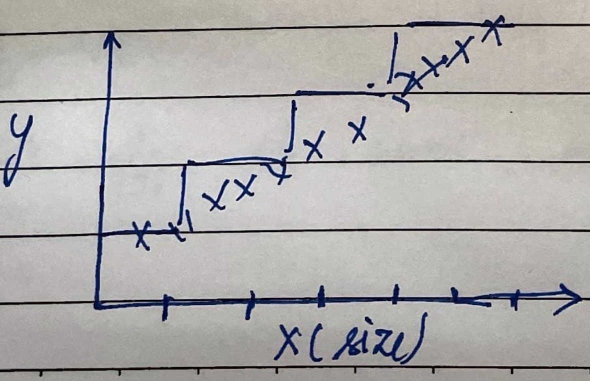
State space $\rightarrow \mathbb{R}^n$

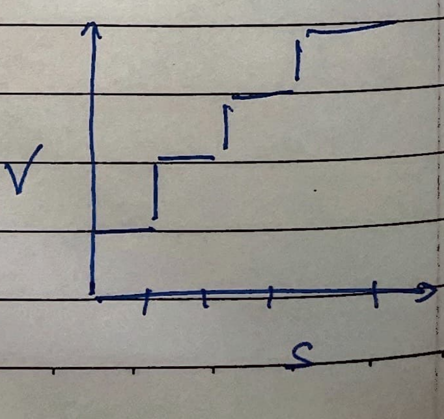
Discretization $\rightarrow$



Problems $\rightarrow$ - Naive representation for V^* (and π^*)

we dont get a very good representation.





curse of dimensionality

$S = \mathbb{R}^n$, and discretize ~~each~~ each dimension in K values, get K^n discrete states.

For 2,3 dimensional state space, discretization would work.

For 4, 6 dimensional state space, discretization would somehow works if done well.

For further dimensions, try not to use discretization.

Alternate approach ⇒ Approximate V^* directly without resorting to discretization

$$y \simeq \theta^T \phi(x) \quad \text{(For linear regression)}$$

$$\phi(x) = \text{features of } x \quad \phi(x) = \begin{bmatrix} x_1 \\ x_2 \\ x_1 \\ x_1 x_2 \end{bmatrix}$$

$$V^*(s) \simeq \theta^T \phi(s)$$

for fitted value iteration

Model simulator of MDP →

(state space) $s \rightarrow$ [Model] $\rightarrow s' \sim P_{sa}$

↑ a (action space)

Car has a state space of 6, but action space of 2 ie steering & brake.

For simplicity, we will now assume that the action space is discrete.

To build the model
- use a physics simulator

$$s' = s + (\Delta t) \cdot \dot{s}$$

$$\Delta t = 0.1 \text{ seconds}$$

- To learn model from data.

$$s_0^{(i)} \xrightarrow{a_0^{(i)}} s_1^{(i)} \xrightarrow{a_1} s_2 \xrightarrow{a_2} s_3 \longrightarrow \cdots \; s_T$$

doing this 'm' times

Now, apply supervised learning to estimate s_{t+1} as function of s_t, a_t

$s_{t+1} \rightarrow s'$ $\qquad s_t \rightarrow s, \quad a_t \rightarrow a$

We can use a linear regression models

$$s_{t+1} = A s_t + B a_t$$

This is a good model for slow speed helicopters.

$$A \rightarrow \mathbb{R}^{n \times n}$$

$$\min_{A,B} \sum_{i=1}^{m} \sum_{t=0}^{T} \left\| s_{t+1}^{(i)} - \left(A s_t^{(i)} + B a_t^{(i)} \right) \right\|^2$$

Model:

$$s_{t+1} = As_t + Ba_t \quad \text{(deterministic)}$$

or

$$s_{t+1} = As_t + Ba_t + \varepsilon_t \quad \text{(stochastic)}$$

where $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$

This is known as model-based RL.

Fitted value iteration - choose feature $\phi(s)$ of some state s

$$V(s) = \theta^T \phi(s)$$

$$\phi(s) = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Going back to value iteration: ~~$V(s) = R(s) + \gamma \gamma$~~

$$V(s) = R(s) + \gamma \max_a \sum_{s'} P_{sa}(s') V(s')$$

$$= R(s) + \gamma \max_a E_{s' \sim P_{sa}} [V(s')]$$

Fitted Value Iteration

~~Sample $\{s^{(1)}, s^{(2)}, \ldots s^{(m)}\} \subseteq S$ randomly~~

~~initiate initialize $\theta := 0$.~~

Using the analogy of linear regression

$X \longrightarrow y$

$s \longrightarrow V(s)$

Fitted Value Iteration

Sample $\{s^{(1)}, s^{(2)}, \ldots s^{(m)}\} \subseteq S$ randomly

Initialize $\theta := 0$

Repeat {

For $i = 1 \ldots m$ {

For each action $a \in A$,

sample $s_1', s_2' \ldots s_K' \sim P_{s^{(i)}, a}$ using model

Set $q(a) = \frac{1}{K} \sum_{j=1}^{K} [R(s^{(i)}) + \gamma V(s_j')]$

So, the value iteration becomes →

$R(s) + \gamma \max_a E_{s' \sim P_a} [V(s')]$

$\downarrow$

$\max_a [E_{s' \sim P_{sa}} [R(s) + \gamma V(s')]]$

Set $y^{(i)} = \max_a q(a)$

In original VI
$V(s^{(i)}) = y^{(i)}$

In Fitted VI
Want $V(s^{(i)}) \simeq y^{(i)}$ (Linear regression)
i.e. $\theta^T \phi(s^{(i)}) \simeq y^{(i)}$

$$\theta = \arg\min_{\theta} \frac{1}{2} \sum_{i=1}^{m} \left(\theta^T \phi(s^{(i)}) - y^{(i)}\right)^2$$

Fitted VI gives approximation to V^*

Implicity defines π^*

$$\pi^*(s) = \arg\max_{a} E_{s' \sim P_{sa}} [V^*(s')]$$

Used sample $s_1', \ldots s_k' \sim P_{sa}$ to approximate expectation.

Say model
$$s_{t+1} = f(s_t, a_t) + \varepsilon_t \quad \text{noise}$$

For deployment / run-time,
set $\varepsilon_t = 0$, and $k = 1$

When in state s
Pick action,

simulator without noise

$$\arg\max_{a} V(\underbrace{f(s, a)})$$

$s' \sim P_{sa}$, but with deterministic simulator.